

A multi-agent framework combining large language models with medical flowcharts for self-triage

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Online health resources and large language models are increasingly used as a first point of contact for medical decision-making, yet their reliability in healthcare remains limited by low accuracy, lack of transparency and susceptibility to unverified information. Here we introduce a proof-of-concept conversational self-triage system that guides large language models with 100 medical flowcharts from the American Medical Association, providing a structured and auditable framework for patient decision support. The system leverages a multi-agent framework consisting of a retrieval agent, a decision agent and a conversation agent to identify the most relevant flowchart, interpret patient responses and deliver personalized, patient-friendly recommendations, respectively. Performance was evaluated at scale using synthetic datasets of simulated conversations. The system achieved 84.1% accuracy in flowchart retrieval ($N = 2,000$) and 99.1% accuracy in flowchart navigation across varied conversational styles and conditions ($N = 37,200$). By combining the flexibility of free-text interaction with the rigour of standardized clinical protocols, this approach demonstrates the feasibility of transparent, accurate and generalizable AI-assisted self-triage, with potential to support informed patient decision-making outside clinical settings while improving healthcare resource utilization.

Self-triage refers to the process by which people assess the urgency of their symptoms and decide when and where to seek medical care. For many people, this decision is made before clinical contact, often under conditions of uncertainty, anxiety and limited medical knowledge. In recent years, consumers have increasingly turned to online resources to support these decisions. According to the National Center for Health Statistics, 58.5% of adults searched for health or medical information online during a 6-month period from July to December 2022¹. Widely used platforms such as WebMD² and Mayo Clinic³ offer a wide range of health content, including symptom

checkers, medication information and educational articles that aim to help users interpret symptoms and determine appropriate next steps. More recently, conversational artificial intelligence systems based on large language models (LLMs) have emerged as an accessible source of health-related guidance following the public release of ChatGPT in 2022. A US online survey of 2,002 participants found that 95.6% of the respondents used search engines for health queries, while 32.6% reported using LLMs such as ChatGPT to address medical or health questions in the year 2024⁴. These trends indicate that for many consumers, the initial point of contact in healthcare has shifted

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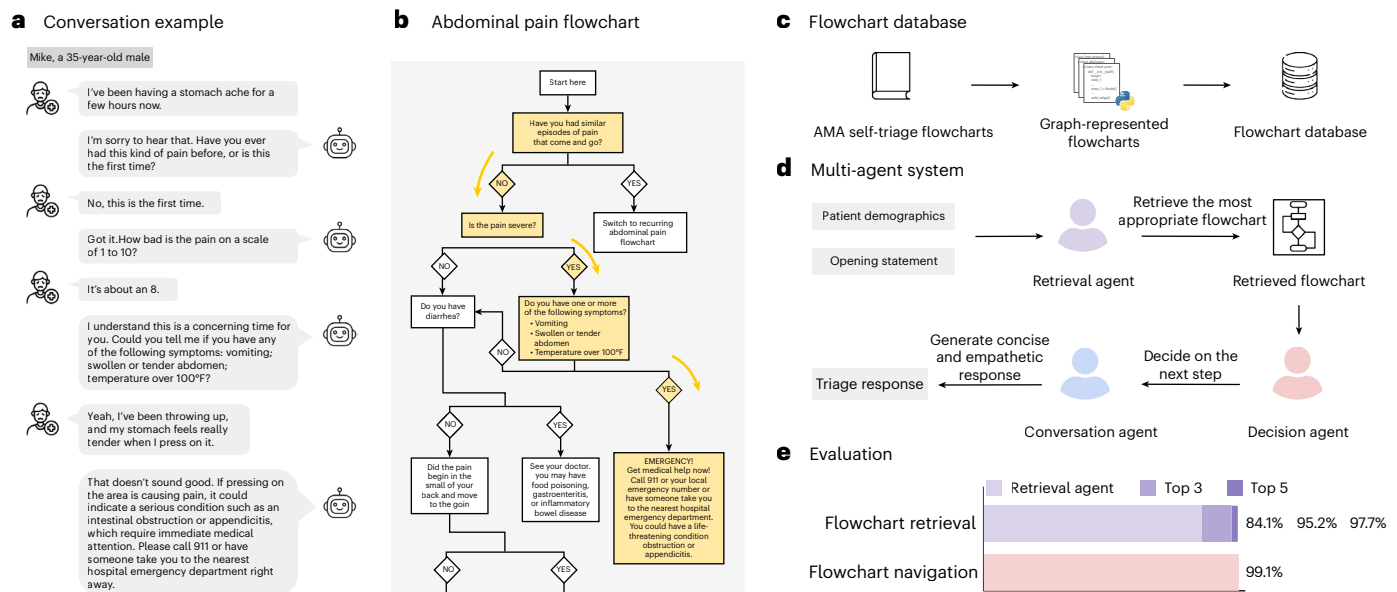


Fig. 1 | Multi-agent self-triage system overview. **a**, An example conversation between a 35-year-old male patient and the TriageMD chatbot regarding abdominal pain. **b**, An example AMA self-triage flowchart for abdominal pain in adults. **c**, The implementation of the flowchart database. We took 100 AMA self-triage flowcharts and converted them to graph representation to create the flowchart database. **d**, The multi-agent system comprises three components: a retrieval agent that selects the most appropriate flowchart from the database on the basis of patient demographics (for example, 35-year-old male) and main

concern (for example, 'I've been having a stomach ache for a few hours now'), a decision agent that interprets patient responses and determines the next step on the basis of the flowchart and a conversation agent that generates concise and empathetic responses on the basis of both the patient input and the output of the decision agent. **e**, TriageMD was evaluated on two tasks, achieving 84.1% accuracy in flowchart retrieval ($N = 2,000$) and 99.1% accuracy in flowchart navigation ($N = 37,200$). Credit: icons, [Flaticon.com](https://www.flaticon.com).

from clinicians to online platforms that promise immediate, low-cost and widely accessible support.

At the same time, the lack of reliable guidance at this early decision stage has meaningful consequences for both individuals and healthcare systems. Many patients choose to visit emergency departments when their conditions are not urgent, while others delay care owing to uncertainty or false reassurance. Nearly 40% of emergency department visits in the USA are classified as medically non-urgent⁵, suggesting that a substantial portion of patients could be more appropriately managed through primary care or walk-in clinics. Such unnecessary emergency department visits not only result in longer waiting time for patients truly requiring emergency intervention but also contribute to physician fatigue and burnout. An effective and accessible self-triage tool could enable patients to receive timely triage recommendations, allowing them to better manage their conditions and plan for clinic visits. Meanwhile, it could support more appropriate use of healthcare resources.

Despite this need, existing self-triage tools have important limitations. While symptom checkers^{6–8} can provide a list of possible causes or conditions, they often suffer from low diagnostic accuracy, require a certain level of health literacy and fail to offer definitive actions tailored to individual cases^{9,10}. Conversational LLM chatbots have improved accessibility and usability through natural language interaction, but they frequently produce vague or overly general advice. These systems are also susceptible to hallucinations, in which incorrect or misleading information is generated as a result of probabilistic text prediction^{11–13}. For users seeking reassurance or actions during moments of concern, such behaviour can increase confusion or erode trust. Moreover, LLMs are considered a 'black box'^{14,15} given their generation process is rarely transparent, making it difficult for users or clinicians to understand or validate the guidance provided^{16,17}. While fine tuning on medical data can improve performance, it is still difficult for healthcare professionals to control or audit the output of LLMs before deploying them for public use. The lack of a shared, structured representation of medically

informed procedures between human experts and LLMs limits the safety and reliability of current consumer-facing solutions.

To address these challenges, we propose a consumer-facing conversational self-triage system that is explicitly guided by clinician-validated triage flowcharts. We present TriageMD, a system that uses American Medical Association (AMA) flowcharts as a shared representation between healthcare practitioners and LLMs. From the user's perspective, the system conducts a natural language conversation that mirrors the step-by-step questioning process used by clinical nurses during triage. The flowchart structure enables the system to ask clinically relevant questions, interpret user responses and guide individuals towards clear next actions. This approach combines the free-text conversational capabilities of LLMs with the rigour of standardized clinical protocols, offering a novel solution for delivering personalized and reliable decision support. We leverage prompt engineering techniques and a multi-agent framework to implement this system (Fig. 1). It consists of a flowchart database that stores graph-represented AMA flowcharts (Fig. 1c) and three agents powered by LLMs: a retrieval agent, a decision agent and a conversation agent (Fig. 1d). These agents collaboratively perform the triage task by retrieving an appropriate flowchart on the basis of patient information, interpreting patient responses and providing patient-friendly guidance, respectively. At the end of the triage process, the system delivers an action recommendation (Fig. 1a) tailored to the individual case, derived from navigating through the relevant AMA-validated flowchart (Fig. 1b).

We evaluated the proposed system through two tasks: flowchart retrieval, which measures the system's ability to identify the most appropriate flowchart from the database, and flowchart navigation, which assesses its ability to interpret patient responses and guide the conversation towards a final recommendation. Given the intended use in consumer-facing self-triage, we aimed to evaluate the system's performance at scale to rigorously test the reliability and faithfulness of its outputs. To this end, we used synthetic datasets generated

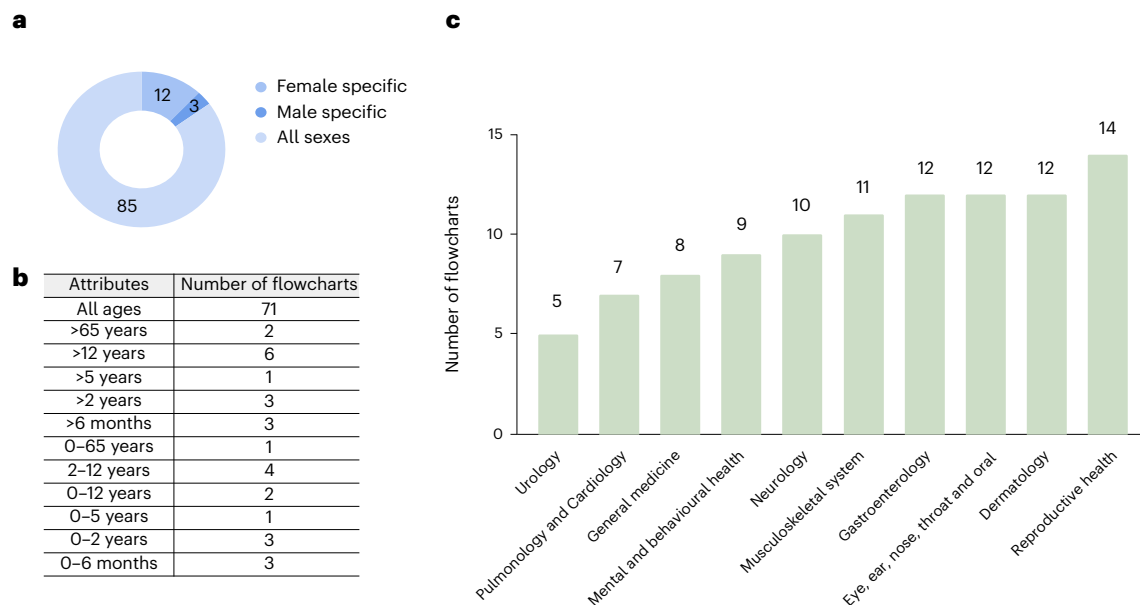


Fig. 2 | Flowchart database breakdown. **a**, The distribution of flowcharts by target sex groups of intended users. **b**, The distribution of flowcharts by target age groups of intended users. **c**, The distribution of flowcharts by medical

specialty. In total, 100 flowcharts were categorized into ten specialties, with urology containing the fewest number of flowcharts ($N = 5$) and reproductive health the most ($N = 14$).

by multiple language models for controlled and repeatable evaluation of simulated conversations. The retrieval results show that the system identified the most appropriate flowchart out of 100 validated AMA flowcharts with an average accuracy of 84.1%, with top-3 accuracy of 95.2% and top-5 accuracy of 97.7%. In the navigation task, the system achieved an average accuracy of 99.1% in interpreting simulated patient responses with diverse conversational styles (that is, brief versus descriptive and clear versus vague) and conditions (that is, yes/no/maybe, off-topic). These findings demonstrate the system's strong potential to deliver consistent and reliable guidance for individuals seeking care decisions, while maintaining transparency through the use of an explicit, human readable and editable database of medical flowcharts.

Results

The self-triage system we developed is structured around three key components: a self-triage flowchart database, a retrieval module that identifies the most relevant flowchart on the basis of patient information and a navigation process that interprets and systematically follows the selected flowchart. To assess the system's performance at scale, which is an essential first step to understand its faithfulness, controllability and failure modes, we utilized large synthetic datasets generated by LLMs. In particular, given the back-and-forth nature of conversational patient triaging, it would not be possible to use a pre-recorded or existing dataset of human conversation to test our AI system. We generated synthetic datasets that enable extensive and controlled testing across diverse clinical scenarios, offering both scalability and comprehensiveness, which is difficult to achieve through human-participant studies. To enhance generalizability, we utilized multiple LLM models across different providers to create datasets for system assessment. As an initial proof-of-concept system, our evaluation focuses on the ability of LLMs to accurately retrieve and execute the proper flowcharts. Specifically, we aim to assess two hypotheses:

- H1: The retrieval module can accurately identify the most appropriate flowchart from the database.
- H2: The navigation module can reliably interpret patient responses and correctly follow the retrieved flowchart.

Flowchart database

We first created a flowchart database of self-triage flowcharts derived from the American Medical Association Family Medical Guide, 4th Edition¹⁸. It covers a wide range of common symptoms and includes flowcharts for particular age or sex groups. We obtained a total of 100 flowcharts after preprocessing (detailed in the Methods) and illustrated the distribution in Fig. 2. Notably, the majority of the flowcharts are applicable for all sexes (Fig. 2a) and ages (Fig. 2b), with subsets targeted towards specific populations, such as female or paediatric patients. Sex-specific flowcharts focus on symptoms involving sex-specific tissues, while age-specific flowcharts address common symptoms adapted for particular age groups. Figure 2c shows the diverse coverage of the flowchart database across medical specialties. Overall, 100 flowcharts were categorized into ten specialties, with urology containing the fewest number of flowcharts ($N = 5$) and reproductive health containing the most ($N = 14$).

Retrieving the most appropriate flowchart from the database

We assessed the system performance in selecting the most appropriate flowchart from the database on the basis of the patient's demographic information and main concern. This selection is achieved by a retrieval agent implemented using retrieval-augmented generation (RAG), as detailed in the Methods. Synthetic patient demographics and opening statements, defined as the first sentence a patient would use to express their main concern, were generated given the flowchart descriptions using four different models: GPT-4o¹⁹, Claude 3 Haiku²⁰, Gemini 2.0 Flash-Lite²¹ and DeepSeek Chat²².

Figure 3 presents the flowchart retrieval result evaluated on the synthetic opening statement dataset. Figure 3a shows the structure of the retrieval agent, which consists of a similarity search module followed by an LLM (we used the OpenAI GPT-4o-mini²³ model for cost efficiency). The patient's demographic information and opening statement are first processed through a cosine similarity search to retrieve the top-ranked flowcharts according to their similarity scores. The retrieved candidates are then passed to the LLM, which selects the most appropriate flowchart on the basis of both the flowchart information and the similarity scores.

The module components in Fig. 3a correspond to the retrieval performance in Fig. 3b based on $N = 2,000$ opening statements comprising

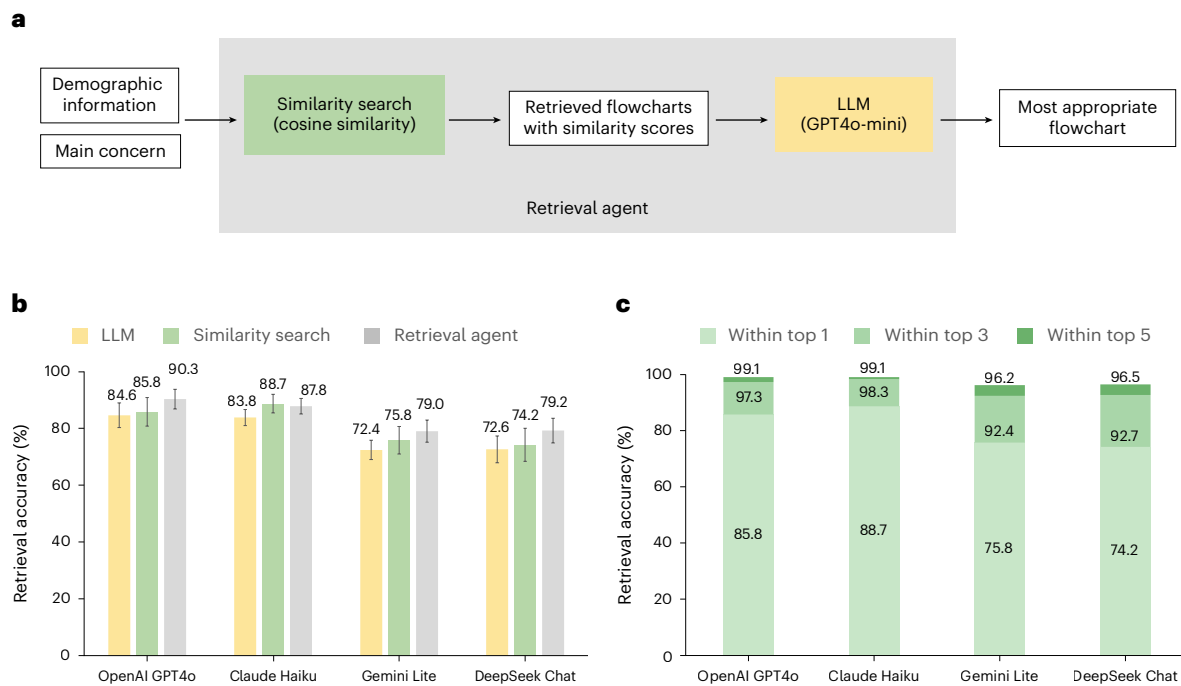


Fig. 3 | Evaluation of flowchart retrieval accuracy on synthetic opening statement data. a, The structure of the retrieval agent, which consists of a similarity search (cosine similarity) module followed by an LLM (GPT-4o-mini). Patient demographic information and opening statements (defined as the first sentence a patient would use to express their main concern) are processed through cosine similarity search to retrieve top-ranked flowcharts with similarity scores, which are then analysed by the LLM to select the most appropriate flowchart. **b,** The retrieval accuracy across LLM-only, similarity search-only (top-1 result) and combined retrieval agent settings, evaluated on opening statements generated by GPT-4o, Claude Haiku, Gemini Lite and DeepSeek Chat. The retrieval agent achieves the highest average accuracy ($84.1\% \pm 5.8\%$) while

showing a comparable performance across different LLM-generated statements. Note, in the retrieval agent setting, the similarity search module returns the top-10 flowchart to be considered by the LLM module. **c,** The retrieval accuracy from top-1, top-3 and top-5 retrieved results. We further extended the similarity search results to the top-3 and top-5. The average accuracies for retrieving the correct flowchart within the top-3 and top-5 retrieved results are $95.2\% \pm 3.1\%$ and $97.7\% \pm 1.6\%$, respectively. This suggests that the similarity search module can effectively retrieve and rank the relevant flowcharts, which supports a more inclusive decision-making process. Data are presented as means $\pm$ s.d. of 2,000 opening statements ($N = 2,000$).

20 per AMA flowchart. When using only the LLM module (Fig. 3, yellow), which relies solely on the LLM's semantic comprehension, the average accuracy of correctly identifying the relevant flowchart is $78.3\% (\pm \text{s.d. } 6.8\%, N = 2,000)$. Similarly, when utilizing only the similarity search module (Fig. 3, green), which mathematically compares distances within vector space, the accuracy of identifying the correct flowchart as the closest is $81.1\% (\pm \text{s.d. } 7.2\%, N = 2,000)$. When the two modules are combined to form the retrieval agent (Fig. 3, grey), the average accuracy improves to $84.1\% (\pm \text{s.d. } 5.8\%, N = 2,000)$. This demonstrates that the retrieval agent not only considers the results from the similarity search (with top- n set to 10) but also performs semantic analysis of each retrieved candidate to make a final selection.

We further analysed retrieval failures and identified three dominant reasons: symptom overlap (42.2% on average), demographic profile influence (40.5% on average) and confusing opening statements (17.3% on average). Symptom overlap accounts for the largest proportion of errors and occurs when an opening statement generated for a given flowchart contains symptoms that could be related to another flowchart, or opening statements that mention multiple clinically related symptoms. For example, chest pain is often accompanied by shortness of breath, leading to high semantic similarity and making it difficult to select a single, most appropriate flowchart. Demographic profile influence reflects when age or sex information strongly biases the retrieval towards flowcharts associated with specific populations. Confusing openings refer to opening statements that include extra details that distract from the main symptom and cause the model to misidentify user intent. Notably, these errors are largely benign, as they tend to reflect ambiguity inherent in patient self-reporting rather than

clinically unsafe outcomes. We provide the complete failure analysis result in the Supplementary Table 1.

In addition, Fig. 3b also presents a comparison of retrieval accuracy across opening statements generated by four different models. In the LLM-only setting, performance on GPT-4o- and Claude Haiku-generated opening statements exhibits similar accuracy: 84.6% and 83.8%, respectively, while slightly lower accuracy is observed on Gemini Lite- and DeepSeek Chat-generated statements at 72.4% and 72.6%. For the similarity search-only setting, accuracy is 85.8% for GPT-4o, 88.7% for Claude Haiku, 75.8% for Gemini Lite and 74.2% for DeepSeek Chat. In the full retrieval agent setting, accuracy reaches 90.3% for GPT-4o, 87.8% for Claude Haiku, 79.0% for Gemini Lite and 79.2% for DeepSeek Chat. A complete result with standard deviation is included in Supplementary Table 5. Additional comparisons are reported in Supplementary Information Section 3 that indicate that the open-sourced model achieves similar performance, while a larger state-of-the-art model yields only modest gains relative to its much higher cost.

Figure 3c specifically evaluates the top results returned from the similarity search component. Here we vary the n value to assess the accuracy of whether the correct flowchart appears within the top- n results, ranked by similarity score. The top-1 result corresponds to the similarity search performance shown in Fig. 3b. When we consider the top-3 retrieved candidates from the similarity search, the average accuracy increases to $95.2\% (\pm \text{s.d. } 3.1\%, N = 2,000)$. Extending this to the top-5 retrieved results further slightly improves the average accuracy to $97.7\% (\pm \text{s.d. } 1.6\%, N = 2,000)$. These findings suggest that the similarity search module can effectively retrieve and rank the relevant flowcharts. In real-world deployment, we expect an interactive system

Table 1 | Five different conversational patterns defined for synthetic patient response generation

Pattern	Definition	Characteristics	Example
Brief	Conclusive and minimalistic: responses that clearly answer the question without additional reasoning, details, or repetition of the question.	1. Brief and direct	Question: 'Are you more than three months pregnant?'
		2. No elaboration or rephrasing the question	Answer: 'Nope.'
		3. Focus on 'yes' or 'no'	
Descriptive	Conclusive and descriptive: responses that clearly answer the question and provide additional details, context or elaboration to support the answer.	1. Clear and definitive	Question: 'Are you more than three months pregnant?'
		2. Expand beyond simply answering the question	Answer: 'No, I just had a test recently, and it's negative.'
		3. Always include details, context or a personal anecdote	
Weak	Vague or partially conclusive: responses that lean towards an answer but include uncertainty or hedge the statement with ambiguous language.	1. Show doubt or ambiguity	Question: 'Are you more than three months pregnant?'
		2. Provide a partial inclination towards 'yes' or 'no' but does not fully commit	Answer: 'I doubt it, but I guess it's possible.'
		3. Indicators such as 'I guess', 'maybe' and 'possibly'	
Uncertain	Inconclusive: responses that remain uncertain owing to a lack of sufficient information, neither confirming nor denying the question.	1. Uncertain	Question: 'Are you more than three months pregnant?'
		2. Indicators such as 'I don't know' or 'I'm not sure'	Answer: 'I'm not sure. I haven't checked yet.'
Off-topic	Irrelevant: responses that are completely unrelated to the question but still make basic conversational sense.	1. Off-topic but plausible	Question: 'Are you more than three months pregnant?'
		2. Introduces unrelated information that does not pertain to the question	Answer: 'Oh, I've been organizing my closet lately. It's such a mess!'

will present both a 'most relevant' flowchart recommendation from the retrieval agent and the top-3 flowchart results from a similarity search as 'potentially relevant' flowcharts. This retrieval process leverages the AI's ability to automatically retrieve from a large database while maintaining transparency to allow patients to stay in the loop during the triage process.

Navigating through the retrieved flowchart

To provide proper guidance to patients, it also requires the system to correctly navigate through the retrieved flowchart in a step-by-step conversational manner. We further evaluated the system's navigation performance using synthetic patient responses generated by the same four language models as aforementioned. Drawing from real-world patient–doctor interaction data^{24–26}, we categorized patient responses to doctors' queries into the following five conversational patterns (see Table 1 for a detailed description and response examples):

- (1) Brief: responses that are short and conclusive, suggesting a clear yes or no.
- (2) Descriptive: responses that are descriptive and conclusive, providing additional details.
- (3) Weak: responses that are vague or ambiguous, but slightly leaning towards yes or no.
- (4) Uncertain: responses that are too inconclusive to be interpreted as a yes or no.
- (5) Off-topic: responses that are completely irrelevant to the question.

These patterns represent diverse patient response styles while preserving conversational realism. We created a large-scale response dataset that included all five patterns, resulting in a total of 37,200 synthetic patient responses across 100 flowcharts (see the Methods for generation details). To assess the navigation performance, the patient response and flowchart node pairs are fed into the decision agent. The decision agent is designed to classify patient responses along four axes: on-topic/off-topic, answered/unanswered, yes/no and certain/uncertain (see Table 2 for details). These axes collectively determine whether the system can confidently advance to the next

Table 2 | Four axes defined for the decision agent's structured output

Axis name	Description
isOnTopic	Return 'yes' if the response is relevant to the question, otherwise return 'no'
isAnswered	Return 'yes' if the response provides a yes or no answer to the question, otherwise return 'no'
actualAnswer	Return 'yes' if the patient answers the question affirmatively, 'no' if the patient answers negatively
isUncertain	Return 'yes' if the response expresses uncertainty (for example, 'maybe', 'not sure' and 'probably'), otherwise return 'no'

node or needs to seek clarification, thus enabling a transparent and informative navigation process.

Figure 4 presents the navigation result based on the decision agent's output (the decision agent was implemented with OpenAI GPT-4o-mini model as the backbone). Overall, the system demonstrated high navigation accuracy across patient responses generated by all four models, achieving an average of 99.1% ($\pm$ s.d. 0.54%, $N = 37,200$), as shown in Fig. 4a. Figure 4b–f provides a detailed breakdown of performance across five conversational patterns. For patterns 1–3, we further break down the result into four categories: (1) certain and correct, (2) uncertain and correct, (3) uncertain and incorrect and (4) certain and incorrect. The system only proceeds when it determines the response is certain and holds at the current position if uncertainty is detected. As such, the 'certain and incorrect' category is the only unacceptable case, as the system misinterprets the patient response and proceeds with the potential of providing false guidance to patients, whereas categories 1, 2 and 3 are all considered acceptable.

For pattern 1 (brief responses), the system achieved an average accuracy of a mean $\pm$ s.d. of $98.11\% \pm 2.13\%$ ($N = 9,300$) across all LLM-generated responses for acceptable decisions (category 1: $96.96\% \pm 2.14\%$, category 2: $0.86\% \pm 0.50\%$ and category 3: $0.29\% \pm 0.32\%$), with $1.89\% \pm 2.13\%$ falling into the unacceptable category (category 4), as shown in Fig. 4b. For pattern 2 (descriptive

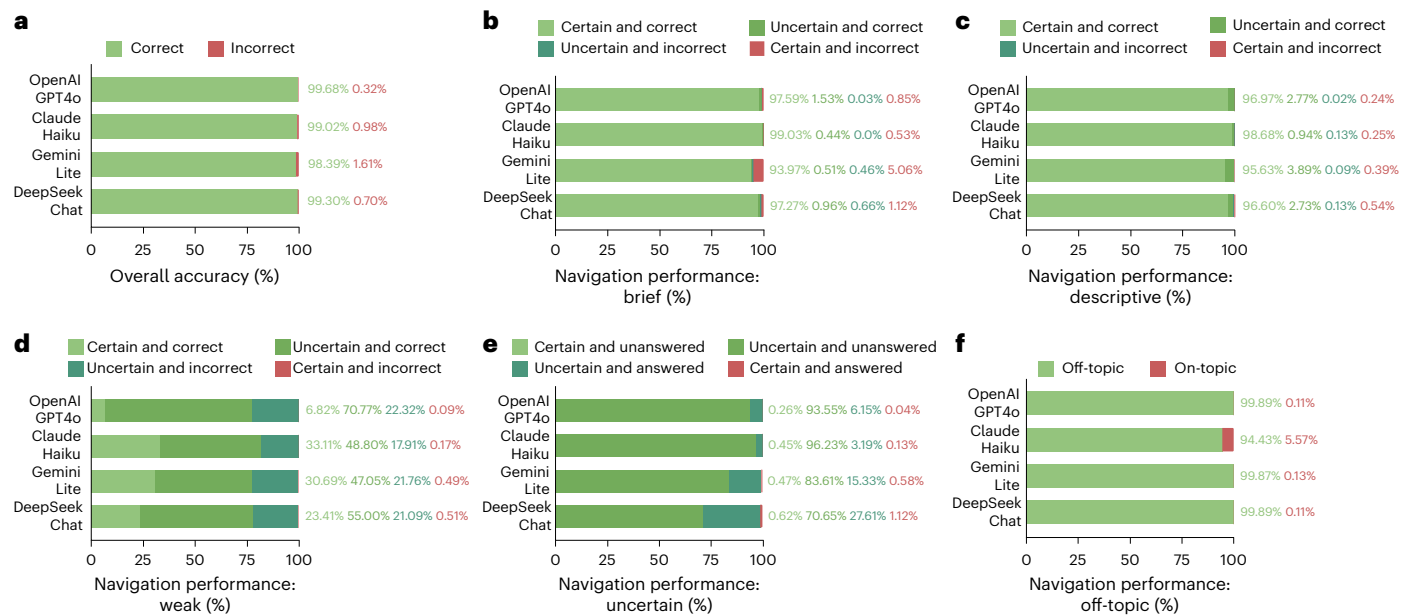


Fig. 4 | Evaluation of flowchart navigation accuracy on synthetic patient response data. **a**, The overall accuracy across patient responses generated by four models. The system achieves 99.10% on average ($\pm$ s.d. 0.54%, $N = 37,200$). **b–f**, The navigation performance breakdown across five conversational patterns: brief (**b**), descriptive (**c**), weak (**d**), uncertain (**e**) and off-topic (**f**) (see Table 1 for detailed pattern explanations and examples). For the first three patterns (**b–d**), we present the results in four categories: certain and correct (light green), uncertain and correct (green), uncertain and incorrect (blue–green) and certain and incorrect (red). Only the last category is unacceptable as the system would not proceed if uncertainty is detected. Acceptable accuracy reached (mean $\pm$ s.d.) 98.11% $\pm$ 2.13%

($N = 9,300$) for brief responses and 99.65% $\pm$ 0.14% ($N = 9,300$) for descriptive responses. For weak responses that are designed to be ambiguous, the acceptable accuracy was 99.69% $\pm$ 0.22% ($N = 9,300$) on average, with the majority classified as uncertain. For uncertain responses in **e**, we slightly modified the category name to reflect their non-answer nature: certain and unanswered (light green), uncertain and unanswered (green), uncertain and answered (blue–green) and certain and answered (red). The system achieved 99.53% $\pm$ 0.49% ($N = 4,650$) accuracy for acceptable categories. Off-topic responses in **f** were correctly identified at 98.52% $\pm$ 2.73% ($N = 4,650$). The system demonstrates robust navigation accuracy across diverse conversational patterns.

responses), the average accuracy for acceptable decisions was 99.65% $\pm$ 0.14% (category 1: 96.97% $\pm$ 1.27%, category 2: 2.58% $\pm$ 1.22% and category 3: 0.09% $\pm$ 0.05%; $N = 9,300$), with only 0.35% $\pm$ 0.14% classified as ‘certain and incorrect (category 4), as illustrated in Fig. 4c. For pattern 3 (weak responses), the system maintained a high performance, with 99.69% $\pm$ 0.22% ($N = 9,300$) acceptable decisions (category 1: 23.51% $\pm$ 11.87%, category 2: 55.41% $\pm$ 10.80% and category 3: 20.77% $\pm$ 1.97%), and 0.31% $\pm$ 0.22% being unacceptable (category 4), as shown in Fig. 4d. This result suggests that even when patient responses were intentionally ambiguous, the TriageMD system was able to detect the underlying uncertainty, as a large portion of the responses is classified as uncertain (categories 2 and 3).

As both pattern 4 (uncertain responses) and pattern 5 (off-topic responses) are designed to be inconclusive, the correct system behaviour is to refrain from proceeding to the next node until a certain, on-topic response is provided. For pattern 4, the navigation process is considered correct if it remains at the current step and seeks confirmation from the patient. The performance for pattern 4 across responses generated by four language models is presented in Fig. 4e. Similarly, the acceptable cases are: (1) certain and unanswered, (2) uncertain and unanswered and (3) uncertain and answered, while the only unacceptable case is (4) certain and answered. The average accuracy for acceptable decisions is 99.53% $\pm$ 0.49% (category 1: 0.45% $\pm$ 0.15%, category 2: 86.01% $\pm$ 11.59% and category 3: 13.07% $\pm$ 10.99%; $N = 4,650$), with only 0.47% $\pm$ 0.49% classified as unacceptable (category 4).

For pattern 5, the navigation process is considered correct if the system classifies the response as off-topic and restates the question. As shown in Fig. 4f, the average accuracy across all LLM-generated patient responses is 98.52% $\pm$ 2.73% ($N = 4,650$), with the highest error rate being 5.57% from Claude Haiku. Overall, the system exhibits high accuracy in flowchart navigation across patient responses generated by different models. It effectively handles varying degrees of

descriptiveness, ambiguity and irrelevance, consistently producing appropriate guidance and maintaining the integrity of the triage protocol. We provide a complete failure analysis for flowchart navigation results in Supplementary Table 3 and complete evaluation results with standard deviation in Supplementary Table 6. In addition, results from alternative system models (OpenAI GPT-5.2, Llama 3.3 70B Instruct Turbo and MedGemma-27B-text-it) are reported in Supplementary Information Section 3, suggesting comparable performance.

Discussion

In this work, we demonstrate how a conversational self-triage tool can support individuals as they make care decisions outside clinical settings by grounding language model responses in clinician-validated flowcharts. Self-triage is a complex procedure that involves information gathering, language comprehension and decision-making. By using AMA self-triage flowcharts as the foundation, the proposed system is able to guide users through a nurse-like questioning process while maintaining conversational flexibility. The system relies on three LLM-powered agents that work together to identify the appropriate triage flowchart, interpret user responses and communicate guidance in a patient-friendly manner. To evaluate system performance, we conducted two key assessments: flowchart retrieval and flowchart navigation. These evaluations measure the system’s ability to accurately select the appropriate flowchart on the basis of patient input and correctly navigate it to provide guidance. The results show that the system can reliably support self-triage interactions with high accuracy, enabling users to receive informative and personalized guidance that reflects established procedures while remaining accessible through natural language interaction. This flowchart-grounded approach addresses a critical ‘interaction gap’ identified in recent randomized controlled trials of LLMs in healthcare. Bean et al.²⁷ showed that while LLMs possess extensive medical knowledge, this does not translate into effective

triage outcomes for the general public because unstructured interactions often fail to elicit key clinical details from users. By constraining conversations using validated flowcharts, our system ensures that the LLM actively gathers necessary context rather than relying on users to volunteer it, thereby mitigating the usability issues inherent in free-text patient–AI conversations^{27–29}.

As a consumer-facing self-triage system, a key contribution is its ability to perform triage according to clearly defined and auditable medical protocols. When individuals experience symptoms at home, they can interact with the system through a conversational interface, describe concerns in their own words, and respond to follow-up questions. The final action recommendations are clearly grounded in AMA self-triage flowcharts and the conversation history. Moreover, the system is adaptable to modern medical practice through clinician-supervised updates to the flowchart database before deployments. In the future, the flowchart database can be further customized to meet the needs of specific specialties, healthcare institutions or patient populations.

Our study has multiple implications to real-world deployments. First, the flowchart retrieval results in Fig. 3 suggest that both the top-3 flowchart results and the final flowchart recommendation from the retrieval agent should be presented to patients (Fig. 3) for a safe real-world deployment. While the retrieval agent achieved the highest accuracy (84.66%) in Fig. 3b, it could be complemented by the top-3 retrieved results from the similarity search (95.29%), as shown in Fig. 3c. Presenting both options allows flexibility in patient interaction: the system can offer a clear recommendation to those seeking guidance, while also enabling patients who prefer a more active role to compare alternatives. This approach supports both system accuracy and greater transparency in the triage process. Alternative designs could be explored in future deployments when use cases become more specific and targeted. For example, the system could simultaneously navigate the top-3 flowcharts until it gathers sufficient information to narrow the options. Alternatively, if the highest-ranked flowchart fails to yield a clear triage decision, the system could sequentially transition to the secondary flowcharts. Such design choices may be tailored to different clinical contexts, patient populations and acceptable risk thresholds.

Second, our flowchart navigation results highlight that a key feature for consumer-facing conversational systems is the ability to handle a wide range of patient response styles, particularly those involving uncertainty. In clinical training, communication skills are a core component^{30,31}, as clinicians need to extract essential information from patient responses with varying levels of clarity, detail and relevance. Weak or uncertain responses present greater interpretation challenges in patient–doctor interactions^{32,33}. This challenge is also reflected in the performance of our consumer-facing system (Fig. 4d,e). A large proportion of responses fall into the uncertain categories, with a higher rate in the uncertain and incorrect category compared with other response patterns. For real-world deployment, the system should be designed to proceed only when a confirmed answer is received, thereby reducing the risk of providing inappropriate action suggestions. Future improvements can also be made to the flowchart design. For example, assigning risk weights to each question in the flowcharts could enable the system to proceed when receiving an uncertain response to a low-risk node. In addition, conditional paths for uncertain responses could be added to flowcharts to address uncertainty more explicitly. Each yes/no question node could have an additional path to specify the desired action when the response is uncertain, such as merging with one of the existing paths, triggering an alternative question or referring the patient to a clinician.

Despite promising initial results, our study has certain limitations. First of all, the proof-of-concept evaluation results are based on synthetic datasets generated by LLMs. While we design the synthetic data to be diverse and realistic to support extensive assessment, it does not capture all aspects of real-world use, including linguistic ambiguity,

idiomatic expressions and conversational unpredictability. Further, the self-triage flowcharts were published in 2011 and may not fully reflect evolving clinical guidance. Future work will require prospective evaluation with real users using updated flowcharts to assess usability, trust and behavioural impact, as well as studies with healthcare organizations to understand how consumer-facing self-triage tools influence care-seeking decisions and healthcare utilization. A pragmatic next step would involve deploying the system on a patient communication platform (that is, MyChart³⁴) and collecting consented interaction logs from a cohort of users seeking triage decision support. The interaction logs could be reviewed by clinicians to assess triage accuracy, and follow-up surveys could be conducted with users to measure clarity of guidance, perceived usefulness and alignment with subsequent care decisions.

Future work could also explore integration with electronic health records. With appropriate patient authorization, access to patient electronic health record data, including medical history, current medications and primary care information, could allow the system to prefetch relevant contextual data, reduce preliminary questions and improve triage efficiency. The system could also support timely escalation by directly alerting clinicians or care teams when needed. Privacy and data security remain critical. Mitigating the risks associated with sensitive data access will require robust data management practices and transparent consent mechanisms. In addition, as we only developed and tested the system in English with text input, further improvement could expand to multilingual or even multimodal input. For example, patients can upload images to help describe their symptoms, and a visual-language model could be developed to further enhance the triage process.

We propose medically validated flowcharts as a shared representation between healthcare practitioners and LLMs, and present a conversational self-triage system with AMA flowcharts. Our findings suggest that such a system has great potential to serve as a front-line triage tool, benefiting both patients and healthcare providers. While further validation on real-world data is required, the framework lays the groundwork for safe, auditable and accessible decision support tools applicable to a wide range of consumer-facing tasks. By prioritizing transparency, adaptability and efficacy, this work shines a light on a new direction for leveraging structured clinical knowledge to implement explainable and trustworthy AI-driven health tools.

Methods

Self-triage flowcharts and preprocess

We analysed and preprocessed all flowcharts from the American Medical Association Family Medical Guide, 4th Edition, published in 2011 by the AMA¹⁸. This guide offers general information on common medical conditions, self-diagnosis flowcharts and actionable health recommendations, aiming to equip individuals and families with essential medical knowledge about their wellbeing outside the clinical setting. The self-diagnosis flowcharts follow a question–answering format, focusing on providing triage actions while suggesting possible diagnoses.

To construct our flowchart database, we implemented a series of preprocessing steps. First, we divided the original infertility flowchart into two separate flowcharts: ‘infertility in men’ and ‘infertility in women’, as the original flowchart contains two entry points for these two population groups. Second, we modified the flowchart content to fit the goal of self-triage. While self-diagnosis attempts to identify specific medical conditions, self-triage emphasizes providing appropriate actions on the basis of symptom urgency. Our modifications focus on ensuring that each flowchart endpoint delivers a clear, actionable recommendation. This included removing the page references that redirect users to other sections of the book after action items and replacing the term ‘self-diagnosis’ with ‘self-triage’ to better reflect the role and goal of the system. An example of flowchart modification is included in Supplementary Information Section 4. Third, there are

three flowcharts that could not be retrieved directly, as their descriptions redirect users to another chart. We mapped them to their respective parent flowcharts to preserve the logical structure and enable future retrieval.

Flowchart representation

Given the decision-tree structure of the AMA flowcharts and prior research indicating that LLMs exhibit better comprehension of code-based formats^{35–37}, we converted the flowcharts into directional graphs using Python. This graph-based representation allows LLMs to more effectively parse the flow of decision-making, enabling accurate step-by-step reasoning and execution. In our graph representation, each step in the flowchart is modelled as a node, and transitions between steps are directed edges. We defined four node types to capture the functional semantics of the flowcharts as follows: (1) 'N' nodes, which poses yes-or-no questions to the user and is the most common node type; (2) 'F' nodes, which denotes nodes that redirect users to a linked flowchart; (3) 'A' nodes, which represents the endpoints of the flowchart, typically an action recommendation (for example, 'see your doctor') or an educational message (for example, 'headaches often occur with fevers'); and (4) 'I' nodes, which provide brief explanations or reassurance preceding an F node. Notably, nodes are connected by directional and conditional edges, where the direction points to the next step, and the condition is either yes or no as the answer to the N node. An example of the graph representation is provided in Supplementary Information Section 4.

Multi-agent self-triage system

Here, we introduce the multi-agent system, which consists of three LLM-based agents working sequentially to perform self-triage: the retrieval agent, the decision agent and the conversation agent. From a preliminary study, we observed that it is challenging for a single model (we used OpenAI GPT4) to perform flowchart retrieval, current step localization and response generation at the same time; therefore, we adapted the multi-agent framework. The system first collects patient information, including sex, age and main concern. This information is utilized to identify the most relevant flowchart from the database by the retrieval agent. After finalizing on a flowchart, the decision agent and the conversation agent collaborate to execute the triage protocol. Specifically, the decision agent interprets and categorizes the patient's answer to the flowchart question, while the conversation agent generates empathetic and concise responses on the basis of the flowchart content. This modular architecture is designed to reflect triage decision-making workflows, with each agent playing a specialized role that mirrors how healthcare professionals gather information, interpret responses and communicate care recommendations.

Patient information gathering. The triage process begins by collecting the basic demographic information of patients, including sex (female/male) and age, with age units specified in months or years, given that the database contains paediatric flowcharts for infants under 1 year old. Following this, the conversation is initiated by TriageMD, prompting the patient to state their main concern (also referred to as chief complaint in clinical settings). For example, a patient might begin with 'I've been having a stomach ache for a few hours now', as shown in Fig. 1. The demographic details and the patient's opening statement are then combined and provided to the retrieval agent, which requires this input to identify the most appropriate flowchart from the database.

The retrieval agent finds the most appropriate flowchart via similarity search. We leverage the RAG framework to effectively match patient concerns with the most appropriate self-triage flowcharts. To build the retrieval component, we first curated the flowchart descriptions from the source book and converted them into embeddings using Facebook AI Similarity Search. Upon receiving a patient's demographic

details and main concern, this information is also vectorized into an embedded query. We then compare it against the embedded flowchart descriptions to identify the top-*n* most relevant candidates using cosine similarity. These candidate flowcharts, along with the patient's query, are then provided to an LLM, which selects the final, most appropriate flowchart. If no suitable flowchart matches the patient's case, 'no flowchart available' would be returned, and the system would prompt the patient to seek medical attention directly, as it is not authorized to provide guidance for conditions not covered by the flowchart database. The system remains at the flowchart retrieval stage when the patient's input is insufficient to identify a specific flowchart, allowing patients to take multiple turns to articulate their main concerns. We include the detailed prompt for creating the retrieval agent in Supplementary Table 7.

Decision agent for interpreting patient's responses. After retrieving the appropriate flowchart, the system begins following it. This involves interpreting patient responses, determining whether patients have answered the flowchart question and navigating through the flowchart to identify the next step. Specifically, at each question node, the decision agent takes in both the node content and the patient's responses as inputs and outputs answers for four axes, as detailed in Table 2. These axes and their descriptions are included in the agent prompt using a structured output parser, which ensures that the output of the decision agent follows a well-defined, structured format (JSON). For example, in the abdominal pain flowchart, the first node asks, 'have you had similar episodes of pain that come and go?' If the patient responds, 'no, this is the first time', the decision agent generates structured output as follows: isOnTopic=Yes, isAnswered=Yes, actualAnswer=No and isUncertain=No. The detailed prompt is included in Supplementary Table 7. Each axis is essential for determining the next step: the system proceeds to the subsequent node only when a certain (isUncertain=No) yes or no answer (actualAnswer=Yes or actualAnswer=No) is obtained. Otherwise, it remains at the current node and attempts to clarify (isOnTopic=No) or confirm (isUncertain=Yes) the patient's response. The decision agent plays a critical role in the system by navigating through flowchart nodes and sending the correct content to the conversation agent.

The conversation agent provides concise and empathetic responses to patients. The conversation agent serves as the final module in the system, responsible for delivering concise and empathetic responses to the patients. It receives the proper node content from the decision agent and generates a reply tailored to the context of the conversation. The conversation agent's prompt is dynamically constructed on the basis of the output of the decision agent, reflecting the patient's previous response. When the patient clearly answers the question, the conversation agent simply conveys the flowchart node content in a patient-friendly manner. If the response is off-topic, the conversation agent gently acknowledges the patient's input and redirects the conversation by restating the original question. In cases where the response indicates uncertainty, the conversation agent attempts to confirm the answer with the patient. This design allows the agent to maintain coherence over multiple dialogue turns while adapting its tone and messaging to the patient's needs.

Evaluation methods

Flowchart retrieval. To evaluate the retrieval accuracy of the flowchart retrieval module, we first created a synthetic patient opening statement dataset using multiple language models. An opening statement is defined as the first sentence a patient would use to express their main concern to a doctor. This statement, along with demographic information, serves as the input for identifying the most appropriate flowchart. For each flowchart, we generated opening statements by providing its flowchart name, description, applicable age range and

targeted sex group to the models. The flowchart name was later used as the ground truth label for assessing retrieval accuracy.

We selected four language models from different providers for a comprehensive analysis: GPT-4o from OpenAI, Claude 3 Haiku from Anthropic, Gemini 2.0 Flash-Lite from Google and DeepSeek Chat from DeepSeek, mainly considering the cost efficiency. Each model generated 20 opening statements per flowchart: 10 brief statements (≤ 25 words) and 10 descriptive statements (≥ 50 words) to reflect a wide range of real-world patient communication styles. In total, 8,000 opening statements were generated for use in the evaluation. We include the prompt for generating patient demographics and opening statements in Supplementary Table 7, along with generated samples from each model in Supplementary Table 2.

Post-generation processing was applied to ensure consistent formatting of demographic information and opening statements, such as harmonizing age units and removing extraneous meta-text, without altering the linguistic content. Notably, for flowcharts targeting paediatric populations (under 12 years old), the opening statements were phrased from a caregiver's perspective (for example, 'My daughter keeps coughing at night. What should I do?'). Each opening statement, along with its associated demographic data, was input into the retrieval agent, which returned the most appropriate flowchart. The retrieved result was then compared with the ground truth to compute retrieval accuracy. We evaluated the sub-component of the retrieval agent both individually and combined, which demonstrates that the complete RAG framework improves flowchart retrieval accuracy, offering greater control and transparency in the triage process.

Flowchart navigation

Navigating through the flowcharts according to the patient's responses is another key procedure in our triage system. Similarly, we first created a patient response dataset using the same models as stated in the above section. To better simulate the real-world conversational patterns from patient–doctor interactions, we proposed five patterns: (1) brief, (2) descriptive, (3) weak, (4) uncertain and (5) off topic. Detailed explanations along with response examples for each pattern is shown in Table 1.

For each question node in the flowcharts, we generated five patient responses per conversational pattern. As the conclusive and partially conclusive patterns require a yes or no answer, five yes and five no responses were generated for each of the first three patterns listed in Table 1. This results in 40 patient responses per question node. Given that 930 question nodes exist across 100 flowcharts, this resulted in 37,200 patient responses per model (note, Claude Haiku refuses to generate responses for very sensitive questions such as 'do you use a vaginal douche or spray or a spermicidal cream or jelly?', which results in 37,100 patient responses from Claude Haiku) and an aggregate of 148,700 responses across the four models included in the dataset. During response generation, the prompt included the flowchart question, specified conversational pattern, its definition and a predefined answer if applicable. A complete version of the prompt is available in Supplementary Table 7, along with the generated samples in Supplementary Table 4. These generated responses were then input into the decision agent to evaluate flowchart navigation performance. From the four-axis output provided by the decision agent, we further computed the accuracy by evaluating whether the suggested action from the decision agent is correct or not. An incorrect case was defined as one in which the decision agent misinterpreted patient responses, leading to inappropriate guidance. This evaluation task measures the robustness of the flowchart navigation process in a controlled and large-scale setting, offering valuable insights into how patient–chatbot interactions can be effectively guided by structured clinical protocols.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The AMA self-triage flowcharts used in this work are published and can be accessed by obtaining the original volume. Owing to copyright restrictions, the complete set of flowcharts cannot be redistributed by the authors. Instead, we provide a representative example in Supplementary Information Section 4 to illustrate the structure and format of the flowcharts. The full synthetic datasets generated for evaluation will not be released publicly, as they could be used to infer the AMA flowchart content. We provide the prompts we used for generating the datasets and samples from the datasets in Supplementary Information Sections 1 and 2 to support reproducibility. Researchers who would like to replicate our study results can obtain a copy of the book to access the flowchart content and may contact us individually to request the full synthetic datasets we generated.

Code availability

The source code is available via GitHub at <https://github.com/digihealthcsd/Multi-agent-self-triage-system>.

References

- Wang, X. & Cohen, R. A. Health information technology use among adults: United States, July–December 2022. *CDC* <https://www.cdc.gov/nchs/products/databriefs/db482.htm> (2023).
- WebMD – Better information. Better health. *WebMD* <https://www.webmd.com/> (2026).
- Mayo Clinic* <https://www.mayoclinic.org/> (2026).
- Mendel, T., Singh, N., Mann, D. M., Wiesenfeld, B. & Nov, O. Laypeople's use of and attitudes toward large language models and search engines for health queries: survey study. *J. Med. Internet Res.* **27**, e64290 (2025).
- Giannouchos, T. V., Ukert, B. & Wright, B. Concordance in medical urgency classification of discharge diagnoses and reasons for visit. *JAMA Netw. Open* **7**, e2350522 (2024).
- Symptom checker with body from WebMD – check your medical symptoms. *WebMD* <https://symptoms.webmd.com/> (2026).
- Symptom checker. *Mayo Clinic* <https://www.mayoclinic.org/symptom-checker/select-symptom/itt-20009075> (2026).
- Symptom checker | Isabel – the symptom checker doctors use. *Isabel Healthcare Inc.* <https://symptomchecker.isabelhealthcare.com> (2026).
- Aboueid, S., Meyer, S., Wallace, J. R., Mahajan, S. & Chaurasia, A. Young adults' perspectives on the use of symptom checkers for self-triage and self-diagnosis: qualitative study. *JMIR Public Health Surveill.* **7**, e22637 (2021).
- Wallace, W. et al. The diagnostic and triage accuracy of digital and online symptom checker tools: a systematic review. *npj Digit. Med.* **5**, 118 (2022).
- Xu, Z., Jain, S. & Kankanhalli, M. Hallucination is inevitable: an innate limitation of large language models. Preprint at <https://arxiv.org/abs/2401.11817> (2025).
- Lee, P., Bubeck, S. & Petro, J. Benefits, limits, and risks of GPT-4 as an AI chatbot for medicine. *N. Engl. J. Med.* **388**, 1233–1239 (2023).
- Pandit, S. et al. MedHallu: a comprehensive benchmark for detecting medical hallucinations in large language models. In *Proc. of the 2025 Conference on Empirical Methods in Natural Language Processing* (eds Christodoulopoulos, C., Chakraborty, T., Rose, C. & Peng, V.) 2858–2873 (Association for Computational Linguistics, 2025); <https://doi.org/10.18653/v1/2025.emnlp-main.143>.
- Casper, S. et al. Black-box access is insufficient for rigorous AI audits. In *Proc. of the 2024 ACM Conference on Fairness, Accountability, and Transparency* 2254–2272 (Association for Computing Machinery, 2024); <https://doi.org/10.1145/3630106.3659037>.
- Bommasani, R. et al. The 2024 foundation model transparency index. *Trans. Mach. Learn. Res.* <https://openreview.net/forum?id=38cwP8xVxD> (2025).

16. Zhao, H. et al. Explainability for large language models: a survey. *ACM Trans. Intell. Syst. Technol.* **15**, 1–38 (2024).
17. Rosenbacke, R., Melhus, Å, McKee, M. & Stuckler, D. How explainable artificial intelligence can increase or decrease clinicians' trust in AI applications in health care: systematic review. *JMIR AI* **3**, e53207 (2024).
18. American Medical Association. *American Medical Association Family Medical Guide* (Wiley, 2011).
19. GPT-4o System Card. *OpenAI* <https://doi.org/10.48550/arXiv.2410.21276> (2024).
20. Claude 3 Haiku: our fastest model yet. *Anthropic* <https://www.anthropic.com/news/claude-3-haiku> (2024).
21. Gemini 2.0: Flash, Flash-Lite and Pro- Google developers blog. *Google* <https://developers.googleblog.com/en/gemini-2-family-expands/> (2024).
22. DeepSeek-V3 Technical Report. *DeepSeek-AI* <https://doi.org/10.48550/arXiv.2412.19437> (2025).
23. GPT-4o mini. *OpenAI* <https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/> (2024).
24. Collins, L. C., Gablasova, D. & Pill, J. 'Doing questioning' in the emergency department (ED). *Health Commun.* **38**, 2721–2729 (2023).
25. Tai-Seale, M., Stults, C., Zhang, W. & Shumway, M. Expressing uncertainty in clinical interactions between physicians and older patients: what matters?. *Patient Educ. Couns.* **86**, 322–328 (2012).
26. Patel, B., Gheihman, G., Katz, J. T., Begin, A. S. & Solomon, S. R. Navigating uncertainty in clinical practice: a structured approach. *J. Gen. Intern. Med.* **39**, 829–836 (2024).
27. Bean, A. M. et al. Reliability of LLMs as medical assistants for the general public: a randomized preregistered study. *Nat. Med.* **32**, 609–615 (2026).
28. Williams, S. C. et al. A surgical approach to building impactful artificial intelligence. *npj Digit. Med.* **8**, 709 (2025).
29. Milne-Ives, M. et al. The effectiveness of artificial intelligence conversational agents in health care: systematic review. *J. Med. Internet Res.* **22**, e20346 (2020).
30. Iversen, E. D. et al. Communication skills training: a means to promote time-efficient patient-centered communication in clinical practice. *J. Patient-Centered Res. Rev.* **8**, 307–314 (2021).
31. Mata, ÁN. et al. Training in communication skills for self-efficacy of health professionals: a systematic review. *Hum. Resour. Health* **19**, 30 (2021).
32. Swinglehurst, D. & Atkins, S. When 'yes' means 'no': why the small details of clinical interactions matter. *Br. J. Gen. Pract.* **68**, 410–411 (2018).
33. Robinson, J. D. & Heritage, J. Physicians' opening questions and patients' satisfaction. *Patient Educ. Couns.* **60**, 279–285 (2006).
34. Epic systems. *MyChart* <https://www.mychart.org/> (2026).
35. Gao, L. et al. PAL: program-aided language models. In *Proc. of the 40th International Conference on Machine Learning* (eds Krause, A. et al.) 10764–10799 (PMLR, 2023).
36. Mishra, M. et al. Prompting with pseudo-code instructions. In *Proc. of the 2023 Conference on Empirical Methods in Natural Language Processing* (eds Bouamor, H., Pino, J. & Bali, K.) 15178–15197 (Association for Computational Linguistics, 2023); <https://doi.org/10.18653/v1/2023.emnlp-main.939>.
37. Madaan, A., Zhou, S., Alon, U., Yang, Y. & Neubig, G. Language models of code are few-shot commonsense learners. In *Proc. of the 2022 Conference on Empirical Methods in Natural Language Processing* (eds Goldberg, Y., Kozareva, Z. & Zhang, Y.) 1384–1403 (Association for Computational Linguistics, 2022); <https://doi.org/10.18653/v1/2022.emnlp-main.90>.

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Author contributions

Y.L., G.W.C., L.Q., X.L and E.J.W. conceived the work. Y.L., S.Y. and H.J. contributed to the technical implementation. J.W., A.Q., T.L. and M.R. contributed to the clinical insights of this work. Y.L. and S.Y. drafted the original paper. E.J.W. supervised the research. All authors reviewed and approved the paper.

Competing interests

X.L. is an employee of Alphabet and may own stock as part of the standard compensation package. The other authors declare no competing interests.

Additional information

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Software and code

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Data collection	The scripts were implemented using Python, with versions available at: https://github.com/digihealthucsd/Multi-agent-self-triage-system .
Data analysis	All the code is publicly available at: https://github.com/digihealthucsd/Multi-agent-self-triage-system .

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The American Medical Association self-triage flowcharts used in this work are published and can be accessed by obtaining the original volume. Owing to copyright restrictions, the complete set of flowcharts cannot be redistributed by the authors. Instead, we provide a representative example in the Supplementary section 4 to illustrate the structure and format of the flowcharts. The full synthetic datasets generated for evaluation would not be released publicly, as they could be used to infer the AMA flowchart content. We provide the prompts we used for generating the datasets and samples from the datasets in Supplementary section 1 and 2 to

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